22. Bell inequalities with certified local and quantum value brackets

Pure Thought research portfolio

Revised 17 September 2026

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

Two distant experimenters choose measurements and compare their outcomes. If the outcomes were produced only by shared classical instructions, their correlations would obey particular inequalities. Quantum systems can exceed some of those limits. This problem calculates both the classical ceiling and a reliable quantum range for a specified test, so that the size of the difference is established mathematically.

Key terms

- **Bell scenario** — A specification of the parties, their measurement choices and the possible outcomes.
- **Local hidden-variable model** — A model in which outcomes depend on local choices and shared classical information, with measurement-choice independence among the assumptions.
- **Bell inequality** — A linear restriction satisfied by the correlations of such local models.
- **CHSH** — The simplest standard two-party test with two binary measurement choices per party.
- **Quantum strategy** — A specified shared quantum state and local measurements producing correlations.
- **POVM** — A set of positive operators summing to the identity that describes a general quantum measurement.
- **Tsirelson bound** — A quantum upper limit on a Bell expression, with $2\sqrt{2}$ the familiar CHSH value in its standard normalization.
- **NPA hierarchy** — A sequence of matrix-positivity tests providing outer bounds on quantum correlations.
- **Commuting-operator model** — A framework in which different parties' measurements commute; its general correlation set differs from the finite-dimensional tensor-product setting.
- **Facet** — A full boundary face of the appropriate dimension of the classical correlation polytope.

Why this matters

Bell inequalities make the difference between classical local explanations and quantum correlations precise. They also underlie ways to certify quantum behavior from observed input-output relationships. Reliable quantum bounds are essential: a numerical optimization value without a valid upper certificate or achieving strategy may misstate what quantum theory permits.

What progress would mean

A complete small test would include an exact classical maximum, an explicit quantum strategy and a matching or enclosing upper bound. Applications to randomness or cryptography would need additional statistical and security assumptions beyond this mathematical benchmark.

Core question

For a fixed rational Bell functional in a small bipartite scenario, can its exact local value and a certified quantum lower/upper bracket be produced?

Scope and assumptions

Start with two parties, two binary settings each, and CHSH. Fix normalization and no-signaling affine coordinates. The target quantum value is the supremum over finite-dimensional tensor-product strategies; identify its closure when discussing limits. Finite NPA levels provide outer bounds through the commuting-operator framework.

Facet discovery and randomness certification are extensions with separate definitions. Do not equate a finite SDP solution with a realizable strategy.

Mathematical target

For $\beta(P)=\Sigma_abxy\ c_abxyP(ab|xy)$, compute β_L by evaluating every deterministic local strategy. A verified finite-dimensional state and POVMs give $L_Q\leq\beta_Q$. A certified NPA dual or noncommutative SOS gives $\beta_Q\leq U_Q$. Report β_L and $[L_Q,U_Q]$.

The NPA limit characterizes commuting-operator correlations, which need not equal the closure of finite-dimensional tensor-product correlations. Finite-level tightness needs an explicit achieving strategy or a valid extraction theorem.

Required outputs and proof obligations

Export coefficients, local vertices, state/measurements, moment-word list, algebraic relations, dual matrices/SOS, and exact or interval objective bounds. To certify a facet in affine dimension D, verify the inequality for every vertex and show saturating vertices have affine dimension $D-1$. Declare relabeling symmetries when counting classes.

Validation and rejection controls

Check all 16 CHSH deterministic strategies and $\beta_L=2$. Reproduce $2\sqrt{2}$ with an explicit two-qubit strategy and an operator upper-bound identity. Reject an invalid dual PSD matrix or a wrong commutation relation. Distinguish eight CHSH variants from one class under relabeling; positivity facets must be handled separately.

Milestones and research extension

Baseline: exact CHSH lower and upper certificates. Strong: a preselected additional rational functional with a certified bracket, whether or not it closes. Research extension: a new inequality with normalized comparison, or device-independent entropy bounds with an explicit adversary and statistical model.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Navascués, Pironio and Acín, A convergent hierarchy of semidefinite programs — NPA convergence to a commuting-measurement representation.
- Ji et al., MIP*=RE — Strict separation of the relevant correlation sets.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$\beta(P) = \sum_{abxy} c_{abxy} P(ab|xy), \quad \beta_L = \max_{P \text{ deterministic local}} \beta(P), \quad L_Q \le \beta_Q \le U_Q.$$

A finite-dimensional state and measurements certify L_Q; a verified NPA dual or operator SOS certifies U_Q. For CHSH, $\beta_L = 2$ and an achieving two-qubit strategy plus an upper-bound identity give $\beta_Q = 2\sqrt{2}$. NPA's commuting-operator limit must not be identified with the closure of finite-dimensional tensor-product strategies. A facet in affine dimension D needs saturating affine rank $D-1$.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — incorrect convergence target (18, 41, 103–105).** NPA convergence is to commuting measurements. Treating this as general convergence to the finite-dimensional tensor-product set ignores a substantive distinction exposed by MIP*=RE.

2. **Major — optimal strategies (24–27, 909–911).** Moment feasibility alone does not supply states and measurements; extraction requires extra conditions. A constructive lower bound is now mandatory for optimality claims.
3. **Major — facet counts/rank (909, 922).** The eight CHSH variants are not eight inequivalent classes; a D-dimensional polytope facet has affine dimension D−1, so D+1 affinely independent saturating vertices would be impossible for a proper facet.
4. **Major — randomness and "exact" numbers.** Floating-point agreement and solver timing are not certificates. Bell violation alone does not specify a full secure randomness protocol. The replacement focuses on auditable value brackets.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For a fixed rational Bell functional in a small bipartite scenario, can its exact local value and a certified quantum lower/upper bracket be produced?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: exact CHSH lower and upper certificates. Strong: a preselected additional rational functional with a certified bracket, whether or not it closes. Research extension: a new inequality with normalized comparison, or device-independent entropy bounds with an explicit adversary and statistical model.

Sources supporting the corrections

- Navascués, Pironio and Acín, A convergent hierarchy of semidefinite programs — NPA convergence to a commuting-measurement representation.
- Ji et al., MIP*=RE — Strict separation of the relevant correlation sets.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means “not excluded”; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.